

PART A — QUESTIONS

AF15-001

Question:

Solve: $17x - 11 = 91$

- A) 5
- B) 6
- C) 7
- D) 8

AF15-002

Question:

If $15n + 4 = 109$, what is n ?

- A) 6
- B) 7
- C) 8
- D) 9

AF15-003

Question:

Simplify: $8(2x - 3) + 5x$

- A) $16x - 24$
- B) $21x - 24$
- C) $21x - 3$
- D) $13x - 24$

AF15-004

Question:

A company charges \$32 per order plus a \$25 processing fee. Which expression represents the total charge for o orders?

- A) $32o + 25$
- B) $25o + 32$
- C) $57o$
- D) $32 - 25o$

AF15-005

Question:

If $f(x) = 12x - 19$, what is $f(7)$?

- A) 55
- B) 65
- C) 84
- D) 103

AF15-006

Question:

Which ordered pair satisfies the equation $y = 10x - 6$?

- A) (1, 5)
- B) (2, 14)
- C) (3, 25)
- D) (4, 36)

AF15-007

Question:

Solve: $16(x - 4) = 96$

- A) 8
- B) 9
- C) 10
- D) 11

AF15-008

Question:

A pattern begins: 5, 19, 33, 47, ... What is the next number?

- A) 59
- B) 60
- C) 61
- D) 62

AF15-009

Question:

Solve: $x/25 = 4$

- A) 29
- B) 75
- C) 100
- D) 125

AF15-010

Question:

A line has a slope of -12 and crosses the y-axis at 9. Which equation represents the line?

- A) $y = 9x - 12$
- B) $y = -12x + 9$
- C) $y = 12x + 9$
- D) $y = -9x + 12$

AF15-011

Question:

If 9 bins contain 216 parts, how many parts are in 6 bins at the same rate?

- A) 126
- B) 132
- C) 144
- D) 156

AF15-012

Question:

Solve: $120 - 18x = 48$

- A) 3
- B) 4
- C) 5
- D) 6

AF15-013

Question:

Which expression is equivalent to $55a - 24a + 9$?

- A) $31a + 9$
- B) $79a + 9$
- C) $31a - 9$
- D) $55a - 15$

AF15-014

Question:

A shelf starts with 640 items. Workers remove 20 items each hour. Which equation gives the number of items I left after h hours?

- A) $I = 640h - 20$
- B) $I = 20h + 640$
- C) $I = 640 - 20h$
- D) $I = 20h - 640$

AF15-015

Question:

If $y = -14x + 150$, what is y when $x = 9$?

- A) 24
- B) 36
- C) 126
- D) 276

AF15-016

Question:

Solve: $24x - 30 = 15x + 60$

- A) 8
- B) 9
- C) 10
- D) 11

AF15-017

Question:

Which value of x makes $13x - 7$ less than 110?

- A) 8
- B) 9
- C) 10
- D) 11

AF15-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : -7, 8, 23, 38

What is the rule?

- A) $y = 15x - 7$
- B) $y = 7x + 15$
- C) $y = -7x + 15$
- D) $y = 15x + 7$

AF15-019

Question:

Solve: $12(x + 4) = 9x + 75$

- A) 7
- B) 8
- C) 9
- D) 10

AF15-020

Question:

If $a = -10$ and $b = 8$, what is $2a + 3b$?

- A) -44
- B) -4
- C) 4
- D) 44

AF15-021

Question:

Which point is located in Quadrant I?

- A) $(-11, 5)$
- B) $(11, -5)$
- C) $(-11, -5)$
- D) $(11, 5)$

AF15-022

Question:

A worker labels 156 boxes in 12 minutes. At the same rate, how many boxes can the worker label in 7 minutes?

- A) 78
- B) 84
- C) 91
- D) 98

AF15-023

Question:

Solve for t : $N = 20t - 60$

- A) $t = N + 60$
- B) $t = (N + 60)/20$
- C) $t = 20(N + 60)$
- D) $t = N/20 - 60$

AF15-024

Question:

What is the slope of a line that goes through $(3, 8)$ and $(9, 56)$?

- A) 6
- B) 7
- C) 8
- D) 48

AF15-025

Question:

Simplify: $40x - 18 - 29x + 30$

- A) $11x + 12$
- B) $69x + 12$
- C) $11x - 48$
- D) $69x - 48$

AF15-026

Question:

If $g(x) = x^2 + 5x$, what is $g(10)$?

- A) 100
- B) 125
- C) 150
- D) 200

AF15-027

Question:

A sequence follows the rule “add 10, then multiply by 2.” If the first term is 8, what is the second term?

- A) 18
- B) 26
- C) 36
- D) 40

AF15-028

Question:

Solve: $15(x - 3) + 12 = 102$

- A) 8
- B) 9
- C) 10
- D) 11

AF15-029

Question:

A line crosses the y-axis at 16 and falls 8 units for every 1 unit to the right. Which equation represents the line?

- A) $y = 8x + 16$
- B) $y = -8x + 16$
- C) $y = 16x - 8$
- D) $y = -16x + 8$

AF15-030

Question:

If $18/27 = x/81$, what is x ?

- A) 45
- B) 48
- C) 54
- D) 60

AF15-031

Question:

Solve: $-18x + 54 = -126$

- A) -10
- B) -6
- C) 6
- D) 10

AF15-032

Question:

A stockroom starts with 2,000 tags. Each shipment uses 80 tags. Which expression gives the number of tags left after s shipments?

- A) $2,000 + 80s$
- B) $2,000 - 80s$
- C) $80s - 2,000$
- D) $2,000 \div 80s$

AF15-033

Question:

Which equation has $x = 20$ as its solution?

- A) $2x + 12 = 54$
- B) $5x - 18 = 82$
- C) $x/5 + 8 = 13$
- D) $3x - 7 = 50$

PART B — ANSWER KEY + EXPLANATIONS

AF15-001 — Correct: B — Explanation:

Solve $17x - 11 = 91$. Add 11 to both sides: $17x = 102$. Divide by 17: $x = 6$. The most common mistake is subtracting 11 instead of adding 11.

AF15-002 — Correct: B — Explanation:

Start with $15n + 4 = 109$. Subtract 4 from both sides: $15n = 105$. Divide by 15: $n = 7$. A common mistake is stopping at 105 and forgetting to divide by 15.

AF15-003 — Correct: B — Explanation:

Distribute first: $8(2x - 3) = 16x - 24$. Then add $5x$: $16x + 5x - 24 = 21x - 24$. A common mistake is forgetting to multiply -3 by 8.

AF15-004 — Correct: A — Explanation:

The company charges \$32 for each order, so that part is $32o$. The \$25 processing fee is added one time. The total charge is $32o + 25$. A common mistake is treating the processing fee as the per-order charge.

AF15-005 — Correct: B — Explanation:

Substitute 7 for x : $f(7) = 12(7) - 19 = 84 - 19 = 65$. A common mistake is adding 19 instead of subtracting it.

AF15-006 — Correct: B — Explanation:

Test the choices in $y = 10x - 6$. For $(2, 14)$, $10(2) - 6 = 20 - 6 = 14$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF15-007 — Correct: C — Explanation:

Solve $16(x - 4) = 96$. Divide both sides by 16: $x - 4 = 6$. Add 4: $x = 10$. A common mistake is adding 4 before undoing the multiplication by 16.

AF15-008 — Correct: C — Explanation:

The pattern increases by 14 each time: 5, 19, 33, 47. Add 14 to 47: 61. A common mistake is assuming the increase is 15 because the numbers are near multiples of 5.

AF15-009 — Correct: C — Explanation:

$x/25 = 4$ means x divided by 25 equals 4. Multiply both sides by 25: $x = 100$. A common mistake is adding 25 and 4 instead of multiplying.

AF15-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -12 and the y-intercept is 9, so the equation is $y = -12x + 9$. A common mistake is switching the slope and y-intercept.

AF15-011 — Correct: C — Explanation:

Find the unit rate: $216 \text{ parts} \div 9 \text{ bins} = 24 \text{ parts per bin}$. For 6 bins: $24 \times 6 = 144$. A common mistake is multiplying 216 by 6 without finding the rate first.

AF15-012 — Correct: B — Explanation:

Solve $120 - 18x = 48$. Subtract 120 from both sides: $-18x = -72$. Divide by -18: $x = 4$. A common mistake is losing the negative sign after subtracting 120.

AF15-013 — Correct: A — Explanation:

Combine like terms: $55a - 24a = 31a$. The constant 9 stays the same. The expression becomes $31a + 9$. A common mistake is combining the constant with the variable terms.

AF15-014 — Correct: C — Explanation:

The shelf starts with 640 items. Workers remove 20 items each hour, so after h hours, $20h$ items are removed. The number left is $I = 640 - 20h$. A common mistake is adding $20h$ even though items are being removed.

AF15-015 — Correct: A — Explanation:

Substitute $x = 9$ into $y = -14x + 150$: $y = -14(9) + 150 = -126 + 150 = 24$. A common mistake is treating $-14(9)$ as positive 126.

AF15-016 — Correct: C — Explanation:

Solve $24x - 30 = 15x + 60$. Subtract $15x$: $9x - 30 = 60$. Add 30: $9x = 90$. Divide by 9: $x = 10$. A common mistake is adding 30 to only one side.

AF15-017 — Correct: A — Explanation:

Solve $13x - 7 < 110$. Add 7: $13x < 117$. Divide by 13: $x < 9$. The only listed value less than 9 is 8. A common mistake is choosing 9, but $13(9) - 7 = 110$, not less than 110.

AF15-018 — Correct: A — Explanation:

The y-values increase by 15 each time x increases by 1, so the slope is 15. When $x = 0$, $y = -7$, so the y-intercept is -7. The rule is $y = 15x - 7$. A common mistake is using +7 instead of -7.

AF15-019 — Correct: C — Explanation:

Distribute: $12x + 48 = 9x + 75$. Subtract $9x$: $3x + 48 = 75$. Subtract 48: $3x = 27$. Divide by 3: $x = 9$. A common mistake is forgetting to multiply 4 by 12.

AF15-020 — Correct: C — Explanation:

Substitute $a = -10$ and $b = 8$: $2a + 3b = 2(-10) + 3(8) = -20 + 24 = 4$. A common mistake is treating $2(-10)$ as positive 20.

AF15-021 — Correct: D — Explanation:

Quadrant I has a positive x-coordinate and a positive y-coordinate. The point (11, 5) fits this description. A common mistake is choosing a point with only one positive coordinate.

AF15-022 — Correct: C — Explanation:

156 boxes in 12 minutes means $156 \div 12 = 13$ boxes per minute. In 7 minutes: $13 \times 7 = 91$. A common mistake is multiplying 156 by 7 without finding the rate first.

AF15-023 — Correct: B — Explanation:

Start with $N = 20t - 60$. Add 60 to both sides: $N + 60 = 20t$. Divide by 20: $t = (N + 60)/20$. A common mistake is dividing N by 20 first and then adding 60.

AF15-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x . From (3, 8) to (9, 56), y increases by 48 and x increases by 6. The slope is $48 \div 6 = 8$. A common mistake is using 48 as the slope without dividing by the change in x .

AF15-025 — Correct: A — Explanation:

Combine like terms: $40x - 29x = 11x$, and $-18 + 30 = 12$. The simplified expression is $11x + 12$. A common mistake is writing $11x - 12$ because the positive 30 is missed.

AF15-026 — Correct: C — Explanation:

Substitute 10 for x : $g(10) = 10^2 + 5(10) = 100 + 50 = 150$. A common mistake is calculating 10^2 as 20 instead of 100.

AF15-027 — Correct: C — Explanation:

Start with 8. Add 10: 18. Then multiply by 2: 36. A common mistake is multiplying first and then adding 10.

AF15-028 — Correct: B — Explanation:

Solve $15(x - 3) + 12 = 102$. Distribute: $15x - 45 + 12 = 102$. Combine: $15x - 33 = 102$. Add 33: $15x = 135$. Divide by 15: $x = 9$. A common mistake is forgetting to combine -45 and $+12$.

AF15-029 — Correct: B — Explanation:

“Falls 8 units for every 1 unit to the right” means the slope is -8 . The line crosses the y -axis at 16, so the equation is $y = -8x + 16$. A common mistake is using a positive slope because the number 8 is given.

AF15-030 — Correct: C — Explanation:

Use the proportion $18/27 = x/81$. Since $27 \times 3 = 81$, multiply 18 by 3: $x = 54$. A common mistake is adding 54 to the numerator because 27 becomes 81.

AF15-031 — Correct: D — Explanation:

Solve $-18x + 54 = -126$. Subtract 54 from both sides: $-18x = -180$. Divide by -18: $x = 10$. A common mistake is giving -10 because both sides contain negative numbers.

AF15-032 — Correct: B — Explanation:

The stockroom starts with 2,000 tags. Each shipment uses 80 tags, so s shipments use $80s$ tags. The number left is $2,000 - 80s$. A common mistake is adding $80s$ even though tags are being used.

AF15-033 — Correct: B — Explanation:

Test $x = 20$. In choice B, $5(20) - 18 = 100 - 18 = 82$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.